

Winding Tree

Protection against illegitimate booking requests

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The current situation in the Winding Tree ecosystem allows anybody to call any instance of Booking API. URLs of these APIs are available to any platform user in the data published about every hotel.

While this setup is very open, it gives an easy attack vector to any malevolent actor. She can scrape the platform for many Booking APIs and flood them with non-sensical requests. Or even requests that make sense but there is no real intention of fulfilling them — thus paralyzing the operations of the hotel.

Protecting an open, decentralized platform from spam and fraud.

Winding Tree's proposal explained

In traditional systems, you would introduce registration for everybody that would like to call the Booking APIs to fix that. However, that system would typically be centralized and the registration process might easily become non-transparent and expensive.

The first precondition is to know who's calling the API. And while we already have that, a free-form text is not enough: It can easily be spoofed, there is room for multiple identities for a single actor, etc.

The identity problem

To explain this in a more precise way, let's give our actors names:

- *Alice* is an OTA (online travel agency)
- *Bob* is running a Booking API
- *Charlie* is an independent crypto-traveller
- *Dan* is a new travelling startup that would like to join Winding Tree
- *Mallory* wants to ruin Bob's business

In the current situation, **Bob** has no way of differentiating between Alice, Charlie, Dan or Mallory and has to blindly trust all of the incoming Booking

requests.

What Bob needs — without prior knowledge of Alice's, Charlie's, Dan's and Mallory's existence — is:

- To be able to tell whom the requests are coming from. Are they from Alice, Charlie, Dan, Mallory, or someone else?
- To be sure that Mallory is **not posing** as Alice, Charlie or Dan
- To decide whether he **trusts** Alice, Charlie, Dan or Mallory
- To block Mallory from sending more malicious requests

This looks like a typical scenario for asymmetric cryptography! If every caller of a Booking API sends their public key along with the message and signs the message with an associated private key, Bob can verify that the message is intact and since the public key is part of the signed message, he can rely on the identity of the sender.

He can therefore reliably tell which requests are coming from Alice, Charlie and Mallory. Moreover, he can see when a new actor, Dan, enters the whole system and starts interacting with Bob.

With this system, Bob can build a history of Booking requests where he can see who and how often is interacting with his API. This covers the first two requirements of Bob — clear knowledge of the caller's identity represented by a public key.

Trust issues

Now enter Dan. He would like to send Booking requests to Bob as well. However, Bob has never heard of Dan and has no idea if Dan is a legit startup or just wants to destroy Bob's business.

So there should be an option for Bob to find out some information about Dan. Two possibilities come to mind:

A registry of actors verified by something or somebody

This sounds straightforward enough. Before sending his first Booking request, Dan goes through a registration process where he can increase the level of his trustworthiness by providing one or more proofs of his legit intentions, such as:

- Stakes some money — it would be expensive to behave as a malevolent actor on a large scale
- Ties his public key to some existing, real-world asset — for example provides a signed message in a publicly available certificate issued by a widely accepted authority
- The registry might be community driven — a malevolent actor would get quickly downvoted

To make the registry itself trustworthy, it of course cannot be centralized and controlled by a single entity that could manipulate with its contents.

A proof of trustworthiness provided by Dan to Bob, based on an activity with other actors that Bob already trusts

Without a registry, Dan has to find another way of building a trusted relationship with Bob. That might require real-world interactions, signed contracts etc. Bob can then transitively provide a guarantee to other Bobs out there, that Dan is a legitimate actor.

All other Bobs could then take into account how trustworthy the original Bob actually is and figure out if they want to trust Dan.

Bob's free will

Ultimately, Bob is a free and responsible man and can make his own decisions about whom he trusts and whom he doesn't. Any registry or proof can only help him decide.

Also, trust is usually not binary, it's probably more of a scale that can change over time. So Bob probably has to have a way of telling the world how is he handling the incoming Booking requests. For instance, under certain conditions, he might require some more guarantees such as payment in advance from the calling actors.

Proposed solution

This is clearly a complex issue with many actors and many variables. As Winding Tree wants to be as open as possible, we cannot enforce or hardcode any limiting rules, the actors themselves or the market should decide.

However, to move forward, we have to start somewhere. That's why we are proposing to create a smart contract *OTA Index* that would serve as a registry of parties interested in sending Booking requests (we will call them OTAs, but it can be Charlie, the crypto-traveller as well).

On top of that, we will come up with a smart contract (or a protocol) to provide levels of trustworthiness of specific public keys. We plan to use the same mechanism for both Hotels (eventually Airlines) and OTAs. These public keys would then be associated with either the Hotel record owner in Hotel index or would be used to sign Booking requests.

As for the booking protocol, we will define multiple levels of trust — untrusted, common, verified and trusted (subject to change) — that every Booking API operator will be able to setup in any way they need. Maybe Booking requests coming from trusted sources will be accepted every time without a doubt, maybe untrusted (or even unsigned) booking requests will not be accepted at all or only with a previous payment.

Surely though, one of the key components in making the decision in which category to put an incoming request will be the OTA Index and the additional trustworthiness smart contract.

We will also allow the Booking API operators to whitelist or blacklist any caller or trust level unconditionally, but this information should also be publicly available via the Booking API.

. . .

There are surely some blanks in this description we will uncover during the implementation phase, but we believe this is a robust enough proposal that does not sacrifice any of the existing benefits of Winding Tree.

Do you have something to add or would you like to ask about more details? Join our [Telegram](#), any technical discussion should go to our [Google Group](#). If you want to contribute, a good place to start is our [GitHub](#).

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